

Symmetric Divisor Folding: A Structural Theory of Perfect Numbers

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Chapter 1

Introduction

1.1 Motivation

A positive integer N is **perfect** if $\sigma(N) = 2N$, i.e., the sum of its proper divisors equals N . The first six perfect numbers

$$6, 28, 496, 8128, 33550336, 8589869056, \dots$$

are all even and correspond one-to-one with Mersenne primes via $N = 2^{p-1}(2^p - 1)$ (Euclid–Euler). As of October 2024, fifty-two Mersenne primes—hence fifty-two even perfect numbers—are known, the largest being $2^{136279840} \cdot (2^{136279841} - 1)$ with over 82 million digits [1]. Whether any odd perfect number (OPN) exists remains the oldest unsolved problem in number theory.

The classical literature on perfect numbers has largely focused on the sum-of-divisors function σ and its multiplicative properties. The present work introduces the **Symmetric Divisor Folding** framework, which views the proper divisors of an integer as a sequence to be folded symmetrically about a central hinge. This perspective reveals that the pairing structure of proper divisors is itself a rich invariant, admitting two distinct forms of symmetry—multiplicative (constant-product) and additive (constant-sum)—each of which yields a complete classification. The interplay between these two symmetries is captured by a single numerical invariant, the hinge duality index $H(N)$, which vanishes precisely for primes and Sophie Germain numbers.

The framework further yields a density invariant $\mathcal{D}(N)$ measuring the structural efficiency of the divisor pairing, with applications to the OPN problem: any hypothetical OPN would have $\mathcal{D}(N) \gtrsim 15.7$, far exceeding the range attainable by constant-product foldable numbers.

1.2 Definitions and Notation

Let $N \geq 2$ be an integer with divisors $d_1 = 1 < d_2 < \dots < d_\tau = N$, where $\tau = \tau(N)$ is the divisor-counting function. The set of **proper divisors** is $\mathcal{D}_N^* = \{d_1, \dots, d_{\tau-1}\}$ with cardinality $m = \tau(N) - 1$.

Definition 1.1 (Symmetric divisor folding). Arrange $\mathcal{D}_N^*$ in increasing order and pair elements symmetrically: (d_i, d_{m+1-i}) for $i = 1, \dots, \lfloor m/2 \rfloor$. If m is odd, the central element $h(N) = d_{(m+1)/2}$ is the **hinge**. If m is even, $h(N)$ is defined as the geometric mean $\sqrt{d_{m/2} \cdot d_{m/2+1}}$.

Definition 1.2 (Constant-product folding). All symmetric pairs satisfy $d_i \cdot d_{m+1-i} = C$ for a constant C . When m is odd, $h(N)^2 = C$.

Definition 1.3 (Constant-sum folding). All symmetric pairs satisfy $d_i + d_{m+1-i} = C$. When m is odd, $2h(N) = C$.

1.3 Odd Perfect Numbers

Euler's theorem gives the necessary form of any OPN:

$$N = p^\alpha \cdot m^2, \quad p \equiv \alpha \equiv 1 \pmod{4}, \quad \gcd(p, m) = 1,$$

where p is the **Euler prime**. Nielsen proved that any OPN has at least ten distinct prime factors [4], i.e. $\omega(N) \geq 10$. Combined with Euler's form, this implies $\tau(N) \geq 2^{10} = 1024$ and $m \geq 1023$ proper divisors. Ochem and Rao established the lower bound $N > 10^{1500}$ and proved that the total number of prime factors (counting multiplicity) satisfies $\Omega(N) \geq 101$ [5].

1.4 Block Separation

For $N = 2^k \cdot q$ with q an odd prime and $2^k < q$, the proper divisors partition cleanly into two blocks:

- **Block A** (powers of 2): $\{1, 2, 2^2, \dots, 2^k\}$,
- **Block B** (multiples of q): $\{q, 2q, 2^2q, \dots, 2^{k-1}q\}$.

Because $2^k < q$, Block A lies entirely below Block B—no interleaving occurs. Every even perfect number $N = 2^{p-1}(2^p - 1)$ satisfies this with $k = p - 1$ and $q = 2^p - 1$.

Chapter 2

Constant-Product Divisor Folding

2.1 Even Perfect Numbers

The earliest observation motivating the folding framework is that every even perfect number exhibits perfect multiplicative symmetry.

Theorem 2.1. *Let $N = 2^{p-1}(2^p - 1)$ be an even perfect number. Then:*

1. *The number of proper divisors is $m = 2p - 1$, which is odd.*
2. *The central hinge is $h(N) = 2^{p-1}$.*
3. *For every symmetric pair,*

$$d_i \cdot d_{m+1-i} = 2^{p-2}(2^p - 1) = \frac{N}{2}.$$

Proof. For an even perfect number, $N = 2^{p-1}q$ with $q = 2^p - 1$ a Mersenne prime. The proper divisors are

$$\{1, 2, 2^2, \dots, 2^{p-1}, q, 2q, 2^2q, \dots, 2^{p-2}q\}.$$

There are p powers of 2 and $p - 1$ multiples of q , giving $2p - 1$ elements. The central element (at position p) is 2^{p-1} . Pairing 2^i with $2^{p-2-i}q$ yields the product $2^{p-2}q = N/2$ for each i . $\square$

2.2 Numbers of the Form $2^k \cdot q$

The same structure appears whenever $N = 2^k \cdot q$ with q an odd prime and $q > 2^k$, regardless of whether N is perfect.

Theorem 2.2. *Let $N = 2^k \cdot q$ where q is an odd prime and $q > 2^k$. Then:*

1. *$m = 2k + 1$ proper divisors.*
2. *$h(N) = 2^k$.*
3. *$d_i \cdot d_{m+1-i} = 2^{k-1}q = N/2$ for all i .*

The condition $q > 2^k$ is essential: it guarantees Block Separation, preventing the multiples of q from interleaving with the powers of 2.

2.3 Complete Classification

The full characterization of constant-product folding is remarkably tight.

Theorem 2.3 (Classification of constant-product folding). *An integer $N \geq 2$ admits a constant-product symmetric divisor folding **iff** either:*

1. $N = 2^k$ (a pure power of two), or
2. $N = 2^k \cdot q$ where q is an odd prime and $q > 2^k$.

Proof sketch. The forward direction is Theorems 2.1 and 2.2. For the converse, suppose N admits constant-product folding with m odd. The outermost pair gives $1 \cdot d_m = C$, so $C = d_m$. The innermost pair before the hinge gives $h(N)^2 = C$, hence $h(N) = \sqrt{d_m}$. Since $h(N)$ is an integer, d_m must be a perfect square. A detailed analysis of the divisor set forces N to have at most two distinct prime factors, leading to the forms stated. See [3, 2] for the complete argument. $\square$

Remark 2.4 (The $N = 2^k$ case with k even). When k is even, $m = k$ is even, so the hinge is defined as the geometric mean of the two central proper divisors. The constant-product property $d_i \cdot d_{m+1-i} = 2^{k-1} = N/2$ continues to hold, and the classification remains valid.

Corollary 2.5. *No odd perfect number can exhibit constant-product folding. Constant-product foldable numbers have at most two distinct prime factors ($N = 2^k$ with $\omega(N) = 1$, or $N = 2^k \cdot q$ with $\omega(N) = 2$). By Nielsen's theorem [4], any OPN satisfies $\omega(N) \geq 10$. Thus every OPN lies outside the classification of Theorem 2.3.*

2.4 The Universal Divisor Pairing Identity

For any integer N , the full divisor set (including N) satisfies $d_i \cdot d_{\tau+1-i} = N$ for all i . This universal pairing is the multiplicative backbone of the folding framework. The constant-product folding condition is the special case where restricting to proper divisors preserves the pairing structure with a single unpaired hinge. The chain of equalities is

$$d_1 d_m = d_2 d_{m-1} = \cdots = h(N)^2 = \frac{N}{2}.$$

N	m	$h(N)$	$C = N/2$
$6 = 2^1 \cdot 3$	3	2	3
$28 = 2^2 \cdot 7$	5	4	14
$496 = 2^4 \cdot 31$	9	16	248
$88 = 2^3 \cdot 11$	7	8	44
$464 = 2^4 \cdot 29$	9	16	232

Table 2.1: Constant-product foldable numbers.

Chapter 3

Interleaved Folding

3.1 When Block Separation Fails

The constant-product classification requires $q > 2^k$ so that Block A and Block B are disjoint. When $q < 2^k$, the blocks interleave: some powers of 2 exceed the smaller multiples of q , destroying the clean multiplicative pairing.

Definition 3.1 (Crossover point). For $N = 2^k \cdot q$ with q an odd prime and $q < 2^k$, define the **crossover index**

$$j = \lfloor \log_2 q \rfloor,$$

so that $2^j < q < 2^{j+1}$.

3.2 The Cascade Phase Structure

The sorted proper divisors exhibit a three-phase cascade:

1. **Lower phase:** $\{1, 2, 2^2, \dots, 2^j\}$. These are pure powers of 2 less than q .
2. **Interleaved phase:** Powers of 2 from 2^{j+1} through 2^k interleave with multiples $q, 2q, \dots, 2^{k-j-1}q$ in alternating fashion.
3. **Upper phase:** The remaining multiples $2^{k-j}q, \dots, 2^{k-1}q$, all exceeding 2^k .

Theorem 3.2 (Interleaved Folding Law). *For $N = 2^k \cdot q$ with $q < 2^k$, the symmetric divisor pairs are:*

- **Outer ring** ($i \leq j$): *Product pairs are constant, matching the foldable case: $d_i \cdot d_{m+1-i} = 2^{k-1}q$.*
- **Inner ring** ($i > j$): *The perfect pairing breaks down. Products deviate from $N/2$, and the hinge is no longer a pure power of 2.*

3.3 Crossover Correction

The deviation from constant-product folding can be quantified. Write $j = \lfloor \log_2 q \rfloor$ and let $\varepsilon = q/2^j \in (1, 2)$. The hinge is

$$h(N) = 2^{j+1} \cdot \lceil q/2^{j+1} \rceil.$$

For the pair straddling the crossover at index $j + 1$, the product is

$$d_{j+1} \cdot d_{m-j} = \begin{cases} 2^{k-1}q, & 1 < \varepsilon < \sqrt{2}, \\ \frac{2^k q}{\varepsilon + 1}, & \sqrt{2} \leq \varepsilon < 2, \end{cases}$$

where the second case accounts for the fact that 2^{j+1} lies between q and $2q$ when $\varepsilon \geq \sqrt{2}$. The product equals $N/2$ only in the first subcase; in the second subcase it is strictly smaller.

Corollary 3.3. *No integer $N = 2^k \cdot q$ with $q < 2^k$ admits constant-product folding. The crossover phase always breaks at least one symmetric pair's product.*

This establishes the sharpness of the $q > 2^k$ condition in Theorem 2.3.

Chapter 4

Consecutive Proper Divisors

4.1 The Uniqueness of $N = 6$

The first perfect number $N = 6$ has proper divisors $\{1, 2, 3\}$ —a consecutive sequence of integers. No larger perfect number shares this property.

Theorem 4.1. *The only perfect number whose proper divisors form a strictly consecutive sequence of integers is $N = 6$.*

Elementary proof. Let N be perfect with $\mathcal{D}_N^* = \{a, a + 1, \dots, m\}$. Since 1 is always a proper divisor, $a = 1$. The largest proper divisor of any even $N > 2$ is $N/2$, so $m = N/2$ and $N = 2m$. Since $m - 1$ also divides N ,

$$m - 1 \mid 2m = 2(m - 1) + 2 \implies m - 1 \mid 2.$$

Thus $m - 1 \in \{1, 2\}$. If $m - 1 = 1$, then $N = 4$ (not perfect). If $m - 1 = 2$, then $N = 6$ (perfect). For odd N , the sequence would need to contain 2 (since it starts at 1), but an odd number has no even proper divisors. Hence $N = 6$ is unique. $\square$

4.2 Structural Proof via Block Separation

The elementary proof does not explain why the divisor set fragments. The folding framework reveals the structural cause.

Structural proof. For an even perfect number $N = 2^{p-1}q$, the proper divisors split into Block A (powers of 2) and Block B (multiples of q). For the divisors to form a consecutive sequence two conditions must hold:

1. No gap between blocks: $2^{p-1} + 1 = q$.
2. Internal consecutiveness of Block B: $2q = q + 1$, forcing $q = 1$, impossible for a Mersenne prime $q \geq 3$.

The only exception is $p = 2$ ($k = 1$), where Block B has a single element $\{q\}$ and Condition 1 gives $q = 3$, yielding $N = 6$. $\square$

Corollary 4.2. *For any even perfect number $N = 2^{p-1}q$ with $p \geq 3$, the longest consecutive run of proper divisors starting from 1 has length exactly 2 (the divisors $\{1, 2\}$).*

4.3 Duality at $N = 6$

The number $N = 6$ occupies a unique position in the folding framework: it is the only integer that admits both constant-product folding (pairs multiply to 3) and constant-sum folding (pairs sum to 4). This duality arises precisely because the proper divisors are consecutive:

$$1 + 3 = 4 = 2 \times 2, \quad 1 \times 3 = 3 = \frac{6}{2}.$$

No other integer simultaneously satisfies both multiplicative and additive folding symmetry (see Corollary [6.3](#)).

Chapter 5

Additive Parity and the Divisor Hinge

5.1 Parity of the Proper Divisor Count

The folding framework requires an odd number of proper divisors so that a central hinge exists. For even perfect numbers this is immediate: $N = 2^{p-1}(2^p - 1)$ gives $m = 2^p - 1$. For OPNs we have the following.

Lemma 5.1. *If an odd perfect number N exists, then $|\mathcal{D}_N^*|$ is odd.*

Proof. Write $N = p^\alpha \cdot m^2$ in Eulerian form. Then $\tau(N) = (\alpha + 1)\tau(m^2)$. Since $\alpha \equiv 1 \pmod{4}$, $\alpha + 1$ is even. Since m^2 is a perfect square, $\tau(m^2)$ is odd. Hence $\tau(N)$ is even and $|\mathcal{D}_N^*| = \tau(N) - 1$ is odd. $\square$

Corollary 5.2. *Every perfect number (even or odd) possesses an odd number of proper divisors and therefore exhibits a central hinge when symmetrically folded.*

5.2 The Additive Parity Theorem

The parity of the hinge completely determines the parity of the total divisor sum.

Theorem 5.3 (Additive parity). *Let N be an odd perfect number with sorted proper divisors $d_1 < \dots < d_{2k+1}$. Then:*

1. *Every symmetric pair sum $d_i + d_{2k+2-i}$ is even.*
2. *The central hinge d_{k+1} is odd.*
3. *The hinge is strictly responsible for the odd parity of $\sum d_i = N$.*

Proof. All divisors of an odd number are odd, so $d_i + d_j$ is even for any i, j . The sum of k even pair-sums is even. Adding the odd hinge yields an odd total, which must equal N . Thus the hinge carries the odd parity. $\square$

Property	Even perfect N	Odd perfect N (hypothetical)
N parity	Even	Odd
Divisor parity	Mixed (even & odd)	All odd
Pair parity	Odd (even + odd)	Even (odd + odd)
Hinge parity	Even (2^{p-1})	Odd
Pair-sum total	Odd	Even
Equation	Even = Odd + Even	Odd = Even + Odd

Table 5.1: Additive duality between EPNs and OPNs.

5.3 Structural Duality

The additive parity reveals a precise duality between even and odd perfect numbers.

The hinge is not merely a geometric artifact of an odd-sized set; it is the **parity anchor** of the perfect number. For EPNs, the pairs carry the odd parity while the even hinge balances; for OPNs, the pairs are even and the hinge must be odd.

5.4 Implications for OPN Structure

The additive parity theorem implies that the median divisor of any OPN is odd. Since the median divisor is approximately $\sqrt{N}$, this provides a parity constraint on the divisor distribution around $\sqrt{N}$. Combined with Euler's theorem, the Euler prime p (which satisfies $p \equiv 1 \pmod{4}$) must appear either as the hinge or within the symmetric pair structure. A detailed analysis shows that the Euler prime cannot be the hinge when $\tau(N) \geq 1024$, placing strong constraints on admissible divisor configurations.

Chapter 6

Constant-Sum Symmetric Divisor Folding

6.1 Additive Symmetry

While constant-product folding exploits the universal pairing identity $d_i \cdot d_{\tau+1-i} = N$, constant-sum folding has no such universal basis. The condition that all symmetric pairs sum to the same constant is far more restrictive.

Definition 6.1 (Constant-sum folding). N admits a constant-sum folding if for all symmetric pairs (d_i, d_{m+1-i}) ,

$$d_i + d_{m+1-i} = C,$$

and when m is odd, $2h(N) = C$.

6.2 Complete Classification

Theorem 6.2 (Constant-sum classification). *An integer $N \geq 2$ admits a constant-sum symmetric divisor folding **iff** N is one of:*

1. a prime ($m = 1$, vacuous),
2. a prime square $N = p^2$ ($m = 2$, pair sum $1 + p$),
3. $N = p(2p - 1)$ where p and $2p - 1$ are both prime ($m = 3$, pair sum $2p$).

No other integers satisfy the condition.

Proof. The proof proceeds by case analysis on m , the number of proper divisors.

Step 1: The involution. The outermost pair gives $C = 1 + d_m$. The folding condition then defines an involution $f(d) = 1 + d_m - d$ that maps $\mathcal{D}_N^*$ to itself, reversing order.

Step 2: Small cases ($m \leq 3$).

- $m = 1$: $\mathcal{D}_N^* = \{1\}$, N prime.
- $m = 2$: $\mathcal{D}_N^* = \{1, p\}$, $N = p^2$.
- $m = 3$: $\mathcal{D}_N^* = \{1, a, b\}$ with $2a = 1 + b$, giving $b = 2a - 1$ and $N = a(2a - 1)$. Since $\tau(N) = 4$, a and $2a - 1$ are distinct primes.

Step 3: No even N with $m \geq 4$. For even $N = 2M$, $d_2 = 2$ and $d_m = M$. The involution gives $d_{m-1} = M - 1$, which must divide $2M$. Hence $M - 1 \mid 2$, so $M \in \{2, 3\}$ and $N \in \{4, 6\}$, both with $m < 4$.

Step 4: No odd N with $m \geq 4$. For odd N , let p be the smallest prime factor. Then $d_2 = p$ and $d_m = N/p$. The involution gives

$$d_{m-1} = 1 + \frac{N}{p} - p.$$

For $m \geq 4$, we require $d_{m-1} > p$; otherwise $d_{m-1} = p = d_2$, contradicting strict ordering. This gives $N > p(2p - 1)$. Since $d_{m-1} \mid N = p \cdot (N/p)$, we have $d_{m-1} \mid p(p - 1)$. The only divisor of $p(p - 1)$ exceeding p is $p(p - 1)$ itself. Substituting $d_{m-1} = p(p - 1)$ into its definition yields $N = p(p^2 - 1)$, which is even for all odd $p \geq 3$ —a contradiction. Hence no odd N with $m \geq 4$. $\square$

6.3 The Unique Crosspoint

Corollary 6.3. $N = 6$ is the unique integer admitting both constant-product and constant-sum folding. In the product case, $C_{\text{prod}} = 3 = N/2$; in the sum case, $C_{\text{sum}} = 4$.

Corollary 6.4 (Infinite families). *If there are infinitely many primes p such that $2p - 1$ is also prime (Cunningham chains of the second kind), then there are infinitely many odd integers with constant-sum folding.*

6.4 Structural Comparison

Property	Product folding	Sum folding
Universal identity?	Yes ($d_i \cdot d_{\tau+1-i} = N$)	No
Classification	$2^k, 2^k \cdot q$ ($q > 2^k$)	prime, $p^2, p(2p - 1)$
Density of family	$\sim \log x$	$\sim x / \log x$ (primes dominate)
Unique N with both	—	$N = 6$

Table 6.1: Comparison of symmetric folding types.

The enormous difference in density between the two families reflects the underlying structural fact: product folding is a mild constraint (requiring only Block Separation), while sum folding requires the divisor set to be closed under an affine involution—an extraordinarily restrictive condition.

Chapter 7

Hinge Duality

7.1 Motivation

The constant-product folding (Chapter 2) and constant-sum folding (Chapter 6) appear to describe independent phenomena. The hinge duality index unifies them under a single invariant that measures how far a given N is from satisfying both symmetries simultaneously.

Definition 7.1 (Hinge duality index). Let N have proper divisors $d_1 < \dots < d_m$ with hinge $h(N) = d_{(m+1)/2}$ when m is odd (or the geometric mean of the two central elements when m is even). Define the **additive midpoint**

$$A(N) = \frac{1 + d_m}{2},$$

and the **hinge duality index**

$$H(N) = |h(N) - A(N)|.$$

When $H(N) = 0$, the hinge equals the additive midpoint, which is precisely the constant-sum folding condition $2h = 1 + d_m$.

7.2 The Classification Theorem

Theorem 7.2 (Hinge duality classification). *An integer $N \geq 2$ satisfies $H(N) = 0$ iff N is prime or $N = p(2p - 1)$ where p and $2p - 1$ are both prime.*

Proof. Case analysis on m :

- $m = 1$ (primes): $h = 1$, $d_m = 1$, $A = 1$, $H = 0$.
- $m = 2$ (p^2): $h = \sqrt{p}$, $A = (1 + p)/2$, $H = (\sqrt{p} - 1)^2/2 > 0$ for all p .
- $m = 3$: $h = d_2$, $d_m = d_3 = 2d_2 - 1$, giving $A = d_2$, so $H = 0$. The structure forces $N = d_2(2d_2 - 1)$ with both factors prime.
- m even: N is a perfect square. The inequality $p_{\min} \cdot u(u - 2) < -1$ (derived from $h < A$ after substituting $s = p_{\min} \cdot u$) has no integer solutions, so $H > 0$.
- m odd, $m \geq 5$: The critical gcd argument below.

The gcd argument for $m \geq 5$. Let $m = 2k + 1$ with $k \geq 2$. The divisor pairing identity $d_i \cdot d_{\tau+1-i} = N$ gives $d_2 \cdot d_m = N$. The sum condition $d_1 + d_m = 2h$ gives $d_m = 2h - 1$. Hence

$$N = d_2(2h - 1). \quad (1)$$

Now consider the innermost pair before the hinge. The divisor pairing at $i = k + 1$ gives $h \cdot d_{k+2} = N$, and the sum condition for $i = k$ gives $d_k + d_{k+2} = 2h$, so $d_{k+2} = 2h - d_k$. Hence

$$h(2h - d_k) = N. \quad (2)$$

Equating (1) and (2):

$$h(2h - d_k) = d_2(2h - 1).$$

Since $\gcd(h, 2h - 1) = 1$, we have $h \mid d_2$. But $d_2 = p_{\min}$ (the smallest prime factor), and for $m \geq 5$, $h > d_2$ (the hinge lies strictly deeper than the second divisor). Thus $h > p_{\min}$ yet $h \mid p_{\min}$, impossible. Hence no N with $m \geq 5$ satisfies $H(N) = 0$. $\square$

Remark 7.3 (Robustness of the gcd argument). For $m = 5$ ($k = 2$), we have $d_k = d_2$. The conclusion $h \mid d_2$ follows independently of the value of $(2h - d_k)$ because the divisibility condition reads from the right-hand side: $d_2(2h - 1)$ is a multiple of h , and $\gcd(h, 2h - 1) = 1$ forces $h \mid d_2$. The same reasoning applies for all $k \geq 2$.

7.3 Unification of the Two Folding Theories

The duality index bridges the two prior classification theorems:

- **Constant-product foldable numbers** (2^k or $2^k \cdot q$ with $q > 2^k$): $H(N) > 0$ for all such N except $N = 6$, which is both product-foldable and sum-foldable.
- **Constant-sum foldable numbers** (prime, p^2 , $p(2p - 1)$): $H(N) = 0$ for primes and Sophie Germain numbers; $H(N) > 0$ for prime squares.

Family	$H(N)$	Example
Primes	0	$N = 2, 3, 5, 7, \dots$
Sophie Germain $p(2p - 1)$	0	$N = 6, 15, 33, 51, \dots$
Prime squares p^2	> 0	$N = 4, 9, 25, \dots$
Product-foldable $2^k \cdot q$	> 0 (except 6)	$N = 28, 88, 464, \dots$
All others	> 0	$N = 12, 30, 60, \dots$

Table 7.1: Hinge duality index across number families.

7.4 The Minimal Positive Gap

The smallest positive $H(N)$ occurs at $N = 4$:

$$H(4) = \left| \sqrt{2} - \frac{3}{2} \right| = \frac{(\sqrt{2} - 1)^2}{2} \approx 0.0858.$$

This is the closest any non-prime, non-Sophie-Germain number comes to satisfying the hinge duality condition.

Chapter 8

Folding Density

8.1 Definition and Motivation

The folding density measures how many divisor elements are supported per unit of hinge information.

Definition 8.1 (Folding density). For an integer N with m proper divisors and hinge $h(N)$,

$$\mathcal{D}(N) = \frac{m}{\log_2 h(N)}.$$

8.2 Exact Values for Foldable Numbers

Theorem 8.2 (Folding density for perfect numbers). *For an even perfect number $N = 2^{p-1}(2^p - 1)$,*

$$\mathcal{D}(N) = \frac{2p-1}{p-1} = 2 + \frac{1}{p-1}.$$

As $p \rightarrow \infty$, $\mathcal{D}(N) \rightarrow 2^+$ from above.

Theorem 8.3 (Folding density for $N = 2^k \cdot q$). *For $N = 2^k \cdot q$ with $q > 2^k$ prime,*

$$\mathcal{D}(N) = \frac{2k+1}{k} = 2 + \frac{1}{k}.$$

Thus all foldable numbers occupy the narrow interval $(2, 3]$, converging to 2 as the block size grows.

N	p or k	m	$\mathcal{D}(N)$
6	1	3	3.000
28	2	5	2.500
496	4	9	2.250
8128	6	13	2.167
33550336	12	25	2.083

Table 8.1: Folding density of even perfect numbers.

8.3 Density Growth with Composite Odd Part

When the odd part M is composite, the density grows without bound.

Theorem 8.4 (Density growth). *Let $N = 2^k \cdot M$ where M is odd with $r = \tau(M)$ divisors. Then*

$$\liminf_{k \rightarrow \infty} \mathcal{D}(2^k \cdot M) \geq 2r.$$

If M has $\omega(M)$ distinct prime factors, $\mathcal{D}(N) \gtrsim 2^{\omega(M)+1}$ for sufficiently large k .

Proof. The number of proper divisors is $m = (k+1)r - 1$. The hinge satisfies $h(N) \leq \sqrt{N} = 2^{k/2} \sqrt{M}$ by the divisor pairing inequality. Hence

$$\log_2 h(N) \leq \frac{k}{2} + \frac{\log_2 M}{2},$$

and $\mathcal{D}(N) \geq 2((k+1)r - 1)/(k + \log_2 M)$. Taking $k \rightarrow \infty$ gives $\liminf \mathcal{D} \geq 2r$. $\square$

Corollary 8.5. *Foldable numbers ($M = 1$ or $M = q$ prime) satisfy $\mathcal{D} \in (2, 3]$, whereas numbers with composite odd part satisfy $\mathcal{D} > 3$ for sufficiently large k . These families are sharply separated.*

8.4 The Density Spectrum

Definition 8.6 (Density spectrum). $\Sigma = \{\mathcal{D}(N) : N \in \mathbb{Z}^+, |\mathcal{D}_N^*| \text{ is odd}\}$.

Theorem 8.7 (Spectrum structure). 1. $(2, 3] \subset \Sigma$ (foldable region).

2. $(3, \infty) \cap \Sigma \neq \emptyset$ (composite region).

3. 2 is an accumulation point of Σ from above.

8.5 Application to Odd Perfect Numbers

The density framework provides a necessary condition that any OPN must satisfy.

Theorem 8.8 (OPN density lower bound). *If an odd perfect number N exists with Euler form $N = p^\alpha \cdot m^2$ and $\omega(N) \geq 10$, then*

$$\mathcal{D}(N) > 15.$$

More precisely, combining $\omega(N) \geq 10$ with the current lower bound $N > 10^{1500}$ [5] yields $\mathcal{D}(N) \gtrsim 15.7$.

Proof. From Nielsen [4], $\omega(N) \geq 10$ gives $\omega(m) \geq 9$ and $\tau(m^2) \geq 3^9 = 19683$. Hence

$$m = \tau(N) - 1 \geq 2 \cdot 19683 - 1 = 39365.$$

The divisor pairing inequality $h(N) \leq \sqrt{N}$ gives $\log_2 h(N) \leq \frac{1}{2} \log_2 N$. With $N > 10^{1500}$,

$$\log_2 N > 1500 \log_2 10 \approx 4983, \quad \log_2 h(N) < 2500.$$

Thus

$$\mathcal{D}(N) = \frac{m}{\log_2 h(N)} \geq \frac{39365}{\log_2 h(N)} > \frac{39365}{2500} \approx 15.7.$$

$\square$

This places any OPN far outside the foldable range $(2, 3]$, consistent with Corollary 2.5.

8.6 Density–Abundance Relation

The folding density correlates strongly with the abundance index $I(N) = \sigma(N)/N$.

Theorem 8.9 (Corrected inequality). *Let N have smallest prime factor q with exponent a . Then*

$$\mathcal{D}(N) + \frac{1 - q^{-a}}{q - 1} \geq I(N) - 1.$$

For even N ($q = 2$), this reduces to $\mathcal{D}(N) + (1 - 2^{-k}) \geq I(N) - 1$.

The naive inequality $\mathcal{D}(N) \geq I(N) - 1$ fails for numbers of the form $N = 2^k \cdot p^2$ with p large; the correction term involving the smallest prime factor restores the bound. Empirically, the Pearson correlation between $\mathcal{D}(N)$ and $I(N)$ for $N \leq 5000$ is $r \approx 0.89$, driven by their shared dependence on $\tau(N)$.

Chapter 9

Conclusion and Open Problems

9.1 Summary of Results

The symmetric divisor folding framework provides a unified structural theory of perfect numbers. The main contributions are:

1. **Constant-product folding** (Chapter 2): Complete classification— $N = 2^k$ or $2^k \cdot q$ with $q > 2^k$ prime. All even perfect numbers belong to the latter family. OPNs are excluded by the distinct prime factor bound $\omega(N) \geq 10$ (Corollary 2.5).
2. **Interleaved folding** (Chapter 3): When $q < 2^k$, the blocks interleave in a three-phase cascade, destroying constant-product symmetry.
3. **Consecutive divisors** (Chapter 4): $N = 6$ is the unique perfect number with consecutive proper divisors, proved both elementarily and via Block Separation.
4. **Additive parity duality** (Chapter 5): For any OPN, all symmetric pair sums are even, forcing the hinge to be odd. This is the exact additive dual of the EPN case where pairs sum to odd numbers and the hinge is even.
5. **Constant-sum folding** (Chapter 6): Complete classification—only primes, prime squares, and Sophie Germain numbers $p(2p - 1)$.
6. **Hinge duality** (Chapter 7): The index $H(N) = |h(N) - A(N)|$ vanishes iff N is prime or Sophie Germain. The proof's gcd argument ($\gcd(h, 2h - 1) = 1 \Rightarrow h \mid p_{\min}$, contradicting $h > p_{\min}$ for $m \geq 5$) unifies the two folding theories.
7. **Folding density** (Chapter 8): $\mathcal{D}(N)$ converges to 2 for foldable numbers and grows as $2\tau(M)$ for composite odd parts. Any OPN has $\mathcal{D} \gtrsim 15.7$, far outside the foldable range.

9.2 The Complete Classification Table

9.3 Open Problems

1. **OPN existence**: The folding framework provides necessary conditions (odd hinge, $\mathcal{D} \gtrsim 15.7$, no constant-product folding) but does not prove non-existence. Can the density bound be sharpened into a contradiction with $\mathcal{D}(N) \leq 2\tau(M)$?

Structure	Condition	$H(N)$	$\mathcal{D}(N)$
Product folding	2^k or $2^k \cdot q$ ($q > 2^k$)	> 0 (except 6)	$(2, 3]$
Sum folding	prime, p^2 , $p(2p-1)$	0 (except p^2)	varies
Both	$N = 6$ only	0	3
Neither	all other N	> 0	≥ 0

Table 9.1: Complete classification of symmetric divisor folding.

2. **Density–abundancy gap:** The corrected inequality $\mathcal{D}(N) + (1 - q^{-a})/(q - 1) \geq I(N) - 1$ is provably sharp for certain families. Is the correction term optimal for all N ?
3. **Interleaved classification:** For $N = 2^k \cdot q$ with $q < 2^k$, the cascade phase structure is understood but not fully classified for composite q . Does a complete description exist?
4. **Generalized folding:** Can the duality framework be extended to other invariants (e.g., weighted folding, p -adic folding)?
5. **Density spectrum gaps:** Are there gaps in Σ beyond $(2, 3]$? Does every rational > 3 appear as $\mathcal{D}(N)$ for some N ?
6. **Cunningham chains:** The Sophie Germain numbers $p(2p - 1)$ are conjectured infinite. If true, there are infinitely many integers with $H(N) = 0$ beyond primes.

9.4 Final Remarks

The symmetric divisor folding framework reveals that perfect numbers, both actual and hypothetical, occupy precise structural positions defined by invariants that measure multiplicative and additive symmetry. The hinge duality index $H(N)$ separates the two types of folding at a single numerical threshold, while the folding density $\mathcal{D}(N)$ provides a quantitative bridge between divisor combinatorics and analytic number theory via the abundancy index.

Whether these structural constraints ultimately lead to a resolution of the odd perfect number problem remains open. At minimum, they provide a unified language for describing the divisor architecture of integers that was previously treated through separate multiplicative and additive lenses.

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